Temporal Clusters of Words in Conversations Depend on Genre and Frequency

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Conversations are a blend of a few very frequent words, and a vast array of rare words. How do children accomplish learning these rare words?

Although frequency is the best predictor for learning, there is experimental evidence that temporal distribution is an important factor (Schwab and Lew-Williams 2016; Slone et al. 2023). In fact, linguistic units in adult conversations and child-directed speech are often temporally clustered (Altmann, Pierrehumbert, and Motter 2009; Katz 1996; Wojcik and Goulding 2025; Küntay and Slobin 1996; Lester et al. 2022; Waterfall 2006).

Here, we ask whether clustering in child–adult conversations partially compensates for the low frequency of occurrence, explaining why rare words can be learned by children despite their low frequency. In order to answer this question, we need to disentangle clustering from by-chance clustering because of frequency.

Available measures of temporal clustering, such as the dispersion measure in Gries (2008) and Wojcik and Goulding (2025), are correlated with frequency. The burstiness measure in Goh and Barabási (2008) and Slone et al. (2023) works well only for very long observation periods and large amounts of data.

To disentangle frequency from clustering, we introduce a new statistical model. We define appearance as a time-independent, and repetition as a time-dependent process.

The probability of a token to appear in a token-slot is the sum of both processes. There is a constant probability of appearance. Once the token appears, there is a certain probability of repetition.

With increasing distance to the last appearance or repetition, the probability of repetition decreases and eventually drops to the appearance probability. Here we use an exponential function to implement the decay of repetition probability (see Figure 1). The time-dependent repetition probability reflects the degree of temporal clustering independent of frequency.

We applied our model to two genres: adult–child and adult–adult conversations. We want to test whether clustering of rare words differs in the two genres. Our data come from the Manchester Child Language Corpus (Theakston et al. 2001) and the British National Corpus (BNC Consortium 2007).

For comparison, we used tokens that were observed in both genres and occurred at least 20 times. This left us with 801 tokens, 80% of them nouns, verbs, adverbs or adjectives.

We find that in both genres, raw frequency of a token and immediate repetition probability are negatively correlated. Rare tokens are hence in general more often immediately repeated (see Table 1 for regression analysis results).

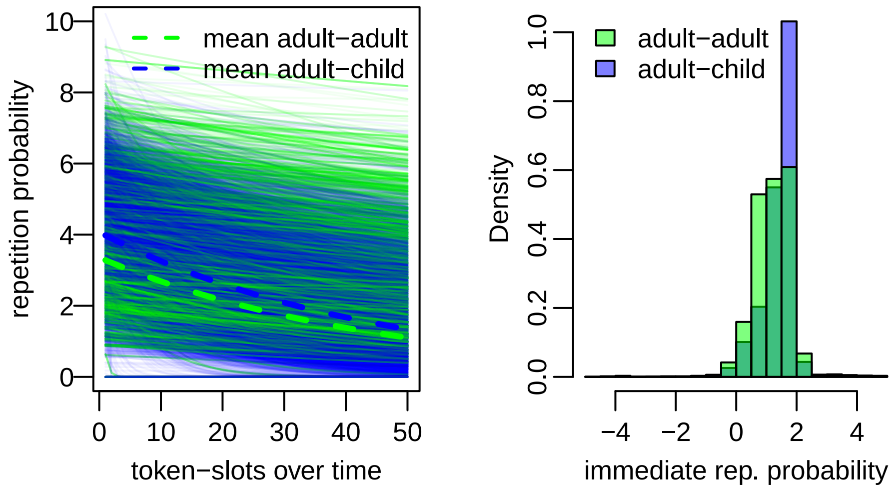

Figure 1: Repetition probabilities of 801 tokens after initial appearance. Left: Repetition probability decrease with increasing distance from initial appearance. Right: logarithm of immediate repetition probabilities at t=1.

We expected that child–adult conversations have a higher immediate repetition probability than adult–adult conversations, but we found that the immediate repetition probability in both genres is similar. We discuss this result in light of the child's contribution to the conversation.

Rare tokens are not immediately repeated more often in the child–adult genre. The general trend for higher immediate repetition probability in rare words compared to more frequent words might still explain how children can also learn rare words, since they are repeated more often, once they appear.

	Estimate	Q2.5	Q97.5
baseline adult–child	-0.51	-0.69	-0.34
adult–adult	0.02	-0.15	0.19
adult–child:log frequency	-0.17	-0.18	-0.16
adult–adult:log frequency	-0.02	-0.02	-0.01
word used by child from the start	0.17	0.14	0.19
word used by child at some point	0.11	0.09	0.13

Table 1: Regression coefficient mean and credible intervals of a model predicting the logarithm of immediate repetition probabilities. The ":" indicates an interaction of predictors. It can be seen that child-adult immediate repetition probability depends on whether the child uses the word as well (rows 5-6). Immediate repetition probability corresponds to the logarithm of the y-axis intercepts in Figure 1. The model was fit with the brms package in R (Bürkner 2021; R Core Team 2024)

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